Virtual Reality Arcade Management System Requirements

The Virtual Reality Arcade Management System is designed to manage the operations of a virtual reality arcade. The arcade serves customers who can reserve VR stations for scheduled gaming sessions. Each customer has a customer ID, first name, last name, phone number and email address. A customer may make bookings, but each booking is made by exactly one customer.

The arcade has multiple VR stations. Each VR station has a station ID, station name or number, equipment type and status. A VR station can be reserved through bookings over time but each booking reserves exactly one VR station. Each booking is a scheduled gaming session. Each booking records a unique booking ID, booking date start time, duration, number of players and booking status.

The arcade offers a few virtual reality games. Each game has a game ID, title, genre, age rating and maximum number of players. A booking can include one or more games, and the same game may appear in different bookings.

I see that employees work at the arcade and help customers and manage scheduled VR sessions. Each employee has an employee ID, first name, last name, job title, phone number and hire date. Each booking is managed by one employee, and an employee may manage many bookings.

The customers make payments for their bookings. Each payment has a payment ID, amount, payment date and payment method. Each booking has one payment, and each payment belongs to exactly one booking.

The system records maintenance performed on VR stations so damaged or unavailable equipment can be tracked. Each maintenance record has a maintenance ID, maintenance date, description and status. Each maintenance record is linked to one VR station and is performed or recorded by exactly one employee. A VR station can have maintenance records over time, and an employee can perform or record many maintenance activities.

The Virtual Reality Arcade Management System aims to organize customer information bookings, VR stations, games, employees, payments and maintenance records so the arcade can operate efficiently.

Process Documentation

The requirements for the Virtual Reality Arcade Management System were collected through online research of real VR arcades. I researched Sandbox VR and Zero Latency VR to understand how VR arcades handle reservations, gaming sessions, equipment, employees and multiplayer experiences.

Sources: Sandbox VR sandboxvr.com, Zero Latency VR zerolatencyvr.com.

From this research I identified the entities: CUSTOMER, BOOKING, VR_STATION, GAME, EMPLOYEE, PAYMENT and MAINTENANCE. I then determined the relationships between them based on how an arcade operates. For example, customers can make bookings, stations can be reserved many times and bookings can include multiple games. The many-to-many relationship between bookings and games was implemented using the BOOKING_GAME table.

I first created the requirements then drew a Chen ER diagram by hand followed by the UML/EER diagram in MySQL Workbench. Finally, I used the UML design to build the database, tables, keys, relationships and SQL scripts.

Project Scope

The project manages the operations of a VR arcade, including customers, bookings, employees, VR stations, games, payments and equipment maintenance. It focuses on database management than building a full customer website or payment system.

Challenges

The main challenge was determining the relationships and cardinalities between entities. Another challenge was converting the the many-to-many relationship between BOOKING and GAME into the BOOKING_GAME junction table. I also had to make sure the Chen diagram, UML diagram and MySQL database were consistent, with each other.

Project Outcomes

The final project produced a working MySQL database called vr_arcade_db with eight tables. Primary keys, foreign keys, indexes and constraints were successfully implemented. Sample data was successfully inserted. JOIN queries were tested to show information from multiple related tables.